REAL-ESTATE UNDERWRITING · PORTFOLIO CASE STUDY

DFW Rental Underwriting Model

VERDICT: PASS AT 7%

A screen-then-underwrite study of the Dallas–Fort Worth single-family rental market

Ajay Patil · Excel + Power Query · Data: Zillow ZHVI & ZORI (June 2026)

THE HEADLINE. 76110 Fort Worth — the strongest risk-adjusted pick of 166 ranked ZIPs — returns just **1.1% levered IRR** at a 7% mortgage, *below* its own 5.0% all-cash return. Cash flow is negative every year, DSCR never reaches 1.0, and rents would need to be ~23% higher to break even. The disciplined call: **pass** — it clears an 8% levered hurdle only if the price falls ~15% (to ~$185K) at 7%, or ~10% (to ~$196K) with a 5–6% mortgage.

ENTRY CAP	LEVERED IRR	UNLEVERED IRR	EQUITY MULT.	COC · YR 1	DSCR · YR 1
4.3%	1.1%	5.0%	1.10x	–6.1%	0.71

Objective

I built an end-to-end model that underwrites a single-family rental in Dallas–Fort Worth and answers one question honestly: **at today's prices and a 7% mortgage, does buying a rental actually earn a return?** The project fills a specific gap in my portfolio — my Chipotle three-statement FP&A model and Python ML forecasting project don't compute *investment-return math*. This one does: cap rate, NOI, DSCR, cash-on-cash, levered and unlevered IRR, and equity multiple, over a full 10-year hold.

The design has two layers. First it **screens** the entire DFW ZIP universe on rental economics to find the most promising submarket; then it **underwrites** that ZIP in depth — a 10-year pro forma, a 360-month amortization schedule, two sensitivity tables, and a cap-rate exit. Built in a single tool (Excel + Power Query) by design, with one firm constraint throughout: **no manufactured returns.** If the honest answer is "pass," that is the finding.

PROJECT SCOPE	
Market	**Dallas–Fort Worth–Arlington**
Asset	**Single-family rental**
Data	**Zillow ZHVI + ZORI (Jun 2026)**
Hold period	**10 years**
Selected ZIP	**76110 Fort Worth**

SKILLS DEMONSTRATED	
Core skill	**Real-estate underwriting**
Modeling	**10-yr pro forma + amortization**
ETL	**Power Query (M)**
Analysis	**cap · DSCR · CoC · IRR · eq. mult.**
Stress testing	**Data Tables + Goal Seek**

What I Built — Methodology

The data funnel is itself a finding — rent coverage is thinner than price coverage:

254 ZIPs with home values (ZHVI) → **220** ZIPs with rents (ZORI) → **216** with BOTH — screenable → **166** full 3-yr rent history — ranked

1 Data & ETL (Power Query)

Loaded two national Zillow research files — home values (ZHVI) and rents (ZORI) — straight from CSV, filtered to the Dallas–Fort Worth–Arlington metro, and reshaped each wide monthly series into a tidy long table (ZIP · City · County · Date · Value) via an unpivot. ZIP codes kept as text to preserve leading zeros.

2 Submarket Screen — and a deliberate override

Ranked the 166 ZIPs with a full 3-year rent history on two independent signals — gross rental yield and 3-year rent CAGR (price-to-rent is the reciprocal of yield) — into a composite weighted 70% yield / 30% growth. The top of the list looked like this:

#	ZIP	City	Price	Rent/mo	Yield	3-yr CAGR
1	75216	Dallas	$177,032	$1,823	12.4%	4.5%
2	75215	Dallas	$177,515	$1,621	11.0%	3.4%
3	75134	Lancaster	$262,549	$2,251	10.3%	5.2%
4	**76110**	**Fort Worth**	**$217,756**	**$1,744**	**9.6%**	**6.2%**
5	75217	Dallas	$201,984	$1,769	10.5%	2.9%
6	75241	Dallas	$215,576	$1,710	9.5%	1.8%

The raw #1–#3 were low-cost South Dallas / Lancaster ZIPs posting 10–12% “yields” — the classic profile of distressed-area risk the data can’t see (condition, taxes, tenant/vacancy risk). **I stepped past them and selected 76110 (rank #4):** a solid 9.6% yield paired with the shortlist’s strongest rent growth, in a conventional, defensible submarket. Knowing when *not* to trust the model’s top output is the analytical point.

3 Underwriting Engine

A named-input assumptions block drives a 360-month amortization schedule (fully paying down to $0) and a 10-year pro forma: gross rent grown annually → vacancy → effective gross income → five escalating operating-expense lines (tax, insurance, management, maintenance, CapEx reserve) → NOI → debt service → cash flow before tax. The exit is a terminal cap rate on forward NOI, net of selling costs and loan payoff. Outputs: cap rate, DSCR, cash-on-cash, equity multiple, and levered & unlevered IRR.

4 Stress Testing

Two live Excel Data Tables flex the result — exit cap × rent growth, and price × mortgage rate — and a Goal Seek solves for the rent that would zero out Year-1 cash flow, mapping exactly what would have to be true for the deal to work.

KEY ASSUMPTIONS	
Purchase price	**$217,756**
Down payment	**25% ($54,439)**
Mortgage	**7.0% · 30-yr fixed**
Rent (Yr 1)	**$1,744 / mo**
Rent growth	**3.0% / yr**
Vacancy	**5%**
Property tax	**2.24% — no homestead**
Insurance	**$2,000 / yr ***
Mgmt / Maint / CapEx	**8% / 5% / 5%**
Exit cap · Sell cost	**5.0% (+70bp vs entry) · 7%**

RESULTS (10-YR, LEVERED)	
Entry cap rate	**4.3%**
Levered IRR	**1.1%**
Unlevered IRR	**5.0%**
Equity multiple **	**1.10x**
Cash-on-cash (Yr 1)	**–6.1%**
DSCR (Yr 1 → 10)	**0.71 → 0.93**
Yr-1 cash flow	**–$3,718**
Cash flow, Yrs 1–10	**negative every yr**
Net sale proceeds	**$92,838**
Breakeven rent	**~$2,144 / mo**

* Insurance is a base-case placeholder below Fort Worth market averages (~$5,500/yr for $300K of dwelling coverage); each +$1,000/yr lowers levered IRR by ~5 points, which only strengthens the pass. ** Equity multiple counts the annual cash shortfalls as additional equity invested.

What the Model Found

Negative leverage. The asset yields ~4.3% unlevered, but the mortgage costs 7%. Financing a 4.3-cap asset with 7% debt *reduces* return — levered IRR (1.1%) lands well below unlevered (5.0%). Side by side, that is the clearest possible proof of the effect.

DSCR below 1.0 for the entire hold. Debt-service coverage runs 0.71 in Year 1 and only reaches 0.93 by Year 10 — NOI never covers the mortgage. Lenders typically require 1.20+, so this wouldn't qualify for standard investment financing at these terms.

Negative carry, all ten years. Year-1 cash flow is –$3,718 and never turns positive (still –$877 in Year 10). The investment requires funding the shortfall out of pocket and relies entirely on appreciation at exit.

Texas property-tax drag. At the full 2.24% non-homestead rate (checked against Tarrant Appraisal District rates), property tax is $4,878 in Year 1 — the largest operating expense (46% of OpEx) and equal to ~52% of NOI. It is the main reason a 9.6% gross yield shrinks to a 4.3% cap rate.

Conservative exit, same answer. The 5.0% exit cap is 70bp above the 4.3% entry cap. Exiting at the entry cap instead lifts levered IRR to only ~5.1% — still well short of an 8% hurdle.

LEVERED IRR SENSITIVITY — PURCHASE PRICE × MORTGAGE RATE

Price ↓ Rate →	5%	6%	7%	8%	9%
$185,093	**13.3%**	**11.7%**	**10.0%**	**8.4%**	6.7%
$195,980	**10.6%**	**8.9%**	7.2%	5.4%	3.6%
$206,868	7.9%	6.1%	4.2%	2.3%	0.3%
$217,756	5.1%	3.1%	**1.1%**	–1.1%	–3.3%
$228,644	2.2%	0.0%	–2.3%	–4.7%	–7.3%
$239,532	–0.9%	–3.5%	–6.1%	–9.0%	–12.0%

Base case (red): $217,756 at 7% = 1.1% levered IRR. Green cells clear an 8% levered hurdle (my target for a leveraged single-family rental) — only at a ~10–15% lower price. Amber = positive but below the hurdle; pink = negative.

Conclusion

Pass at 7%. 76110 is the strongest conventional value the screen surfaced in DFW — and it still doesn't clear the bar. At today's price and rate it delivers roughly a 1% IRR, below risk-free Treasuries, in exchange for leverage, tenant and vacancy risk, illiquidity, and a decade of monthly out-of-pocket carry. Against an 8% levered hurdle, it becomes worth revisiting only if the price falls to ~$185K (–15%) at a 7% mortgage, or to ~$196K (–10%) if rates ease to 5–6%. The value of the exercise isn't a buy recommendation — it's a disciplined, evidence-backed *pass*.

What This Project Demonstrates

End-to-end build. Raw public data → automated ETL → quantitative screen → full underwriting model → stress testing → a written recommendation, all in one reproducible workbook.

Return math. Cap rate, NOI, DSCR, cash-on-cash, levered & unlevered IRR, and equity multiple — the framework used in real-estate and corporate investment analysis.

Judgment over mechanics. Spotting a yield trap in the screen, choosing a forward rent-growth assumption over an unsustainable historical spike, and reaching an honest "pass" rather than forcing a positive number.

Methodology: 10-year pro forma with a 360-month amortization schedule, vacancy, management, CapEx reserve, escalating taxes & insurance, and a terminal-cap exit net of selling costs — all reconciled in a single Excel workbook built with Power Query. Data: Zillow ZHVI (home values) & ZORI (rents, all rental types), ZIP-level, June 2026. Property-tax rate checked against Tarrant Appraisal District rates. · Ajay Patil, MS Finance, The University of Texas at Dallas. · Educational portfolio project — not investment advice.